Example : Program to illustrate Constructor

```
class Sample:                          # class name
    def __init__(self, num):           # Parameter
        print("Constructor of class Sample...")
        self.num=num                   # Instance variable
        print("The value is :", num)
S=Sample(10)
```

The above class "Sample", has only a constructor with one parameter named as num. When the constructor gets executed, first the print statement, prints the "Constructor of class Sample….", then, the passing value to the constructor is assigned to instance variable self.num = num and finally it prints the value passed along with the given string.

The above constructor gets executed automatically, when an object **S** is created with actual parameter 10. Thus, the Python display the following output.

Constructor of class Sample...

The value is : 10

Class variable defined within constructor keep count of number of objects created with the class.

Note

Instance variables are the variables whose value varies from object to object. For every object, a separate copy of the instance variable will be created. Instance variables are declared inside a method using the self keyword. In the above example, we use constructor to define instance variable.

Example : Program to illustrate class variable to keep count of number of objects created.

```
class Sample:
    num=0                              # class variable
    def __init__(self, var):
        Sample.num+=1
        self.var=var                   # instance variable
        print("The object value is = ", self.var)
        print("The count of object created = ", Sample.num)

S1=Sample(15)
S2=Sample(35)
S3=Sample(45)
```


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